VISUAL ANALYTICS COURSEWORK REPORT (EMATM0066)

Abstract

This report presents a visual analytics project (dashboard), exploring socio-economic and health patterns across England and Wales by using 2011 and 2021 Census data. The main motivation is to understand how various socio-economic patterns such as employment, education, housing and health vary across regions and local authorities and how general health evolved between two census years.

Process initialized by combining multiple Census datasets from 2011 and 2021 Census (4 datasets from 2011 Census and 1 dataset from 2021 Census), standardising geographic fields, creating region-level mappings and deriving percentage-based features such as employment rate, health index, percentage highly educated, percentage healthy and percentage unhealthy. These features were utilized to make an interactive, user-friendly Tableau dashboard (Viz) with three viz pages. The first viz page provides a 2011 socio-economic overview using KPI cards (key performance indicators), distribution charts, comparison based on regions and interactive map. On second viz page, PCA and t-SNE projections created in Python to explore similarity, clustering and variation between local authorities based on socio-economic indicators. The final viz page compares health outcomes between 2011 and 2021 using regional comparison chart, a health distribution view, a choropleth map and Bayesian uncertainty intervals for predicted unhealthy population values.

Overall, the dashboard explores the comparison, spatial exploration and analysis, cluster interpretation and uncertainty-aware prediction by understanding regional socio-economic and health aspects. The results suggest that socio-economic and health patterns vary across regions, and that projection and uncertainty views help reveal similarities and model uncertainty between local authorities.

1. Introduction

Census data is one of the most valuable sources for understanding how people live in different areas [6], [7]. It offers information about population, health, employment, education, housing and various other social conditions. Yet, census data is often vast, complex and not easy to interpret directly from raw tables. Because of this, visual analytics can be helpful as it enables users to explore patterns, compare regions and grasp relationships in a clearer way.

This project examines socio-economic and health patterns across England and Wales using data from the 2011 and 2021 Census. The main purpose is to understand how indicators such as employment, education, housing and self-reported health differ across regions and local authorities. These aspects

matter because health is not only a medical term, it is also linked to broader social and economic circumstances. For instance, areas with varying levels of employment, education or home ownership may also display different health outcomes.

The dashboard created for this project is meant to support exploratory analysis rather than provide a single fixed conclusion. It first presents a 2011 socio-economic overview to show the baseline conditions of different regions. Then it applies PCA and t-SNE projections to see whether local authorities with similar socio-economic profiles form visible groups or clusters. Finally, it compares health outcomes between 2011 and 2021, showing regional changes, spatial patterns and uncertainty in predicted unhealthy population values.

The main research question driving this project is:

How can visual analytics help reveal regional socio-economic patterns, health inequalities and changes in self-reported health across England and Wales between the 2011 and 2021 Census?

By combining data preparation, dimensionality reduction, Bayesian modelling and interactive Tableau visualisation, this project seeks to make complex census data easier to explore and interpret. The final dashboard lets users move from a broad overview of socio-economic conditions to more detailed analysis of clusters, regional health differences and uncertainty.

2. Data Preparation and Abstraction

The data preparation stage was an important part of this project because the original census datasets were provided as separate tables with many raw count-based columns. Before building the dashboard, these datasets had to be cleaned, combined and transformed into features that could support fair comparison between local authorities.

2.1 Data Preparation and cleaning

The project used five main census datasets downloaded from **Nomis**, the official UK service for census and labour market statistics [6], [7]. Four of these datasets came from the 2011 Census and covered general health, economic activity, housing tenure and qualification levels. The fifth dataset came from the 2021 Census and covered general health categories.

Before doing actual analysis, the datasets had to be standardised because the original tables used separate structures and field names. Common geographic fields were renamed, such as geography code to area_code and Rural Urban to rural_urban, so that the datasets could be joined using consistent keys. Then, four 2011 datasets were then merged using date, geography, area code and rural urban (these columns are common across all datasets). This created a combined 2011 dataset with 348 local authority records and 48 columns.

2.2 Feature Engineering

After combining the 2011 data, the next step was to create percentage-based features for both the 2011 and 2021 Census data. This was necessary because the original census values were raw population counts. As local authorities have different population sizes, so comparing them directly using only counts would not be fair.

For health, the analysis used healthy percentage, unhealthy percentage, and a health index. The healthy percentage referred to people who had “very good health” or “good health”, while the unhealthy percentage referred to population who had “bad health” or “very bad health”. The health index gave more weight to better health categories which provides a straightforward overall score for comparing general health across areas.

Additional socio-economic features were created from the economic activity, housing, and qualification datasets. These included employment rate, inactivity rate, full-time employment percentage, self-employment percentage, home ownership rate, private rent percentage, percentage with no qualifications, and percentage highly educated. These features helped turn the raw census tables into comparable analytical variables for regions and local authorities.

The original census tables had convoluted structure because categories were stored in a long-format structure, where category names appeared as rows with population values. To make the data easier to work with Tableau, the tables were cleaned, and the data was reshaped (pivoted) so that health categories, job types, housing types, and qualification levels could be used as separate variables.

After this preparation, the four 2011 Census datasets were combined using common geographic fields. The 2021 health data was then joined with the 2011 master dataset using the area code. This made it possible to analyse the 2011 socio-economic indicators and the 2021 health outcomes together.

2.3 Preparing Projection and Bayesian Modelling Features

For dimensionality reduction, a set of percentage-based socio-economic and health features were used as input variables. These included the percentages for health, employment, inactivity, full-time employment, self-employment, ownership, private rent, no qualification, and highly educated groups. The features were standardised using StandardScaler module so that variables measured on different scales would not dominate the projection. **Principal Component Analysis (PCA)** was then applied to generate two linear projection coordinates, PCA1 and PCA2 [4]. The first two PCA components explained around 77.7% of the variance in the selected features. **t-SNE** was also applied to produce TSNE1 and TSNE2 results, which provides a second non-linear projection for examining local similarity and clustering among local authorities.

A **Bayesian Ridge Regression model** was applied to estimate missing or uncertain 2021 unhealthy population percentages [5].

The model was trained on rows where the 2021 unhealthy percentage which derived in data preparation step. Out of 348 records, 323 were used for training, while predictions were generated for all 348 records. Prediction intervals were then created using the predicted value and the standard deviation returned by the model.

Note: Raw population counts were not used directly for PCA or Bayesian modelling because local authorities have very different population sizes. If raw counts were used, the analysis would mainly reflect the size of each area rather than the socio-economic or health profile of the population. Therefore, percentage-based features were used.



Fig. 1. Overview of the Tableau data model and dataset relationships

The final outputs prepared for Tableau were final_master_tableau.csv and final_projection.csv. As shown in Figure 1, final_master_tableau.csv was used as the central master dataset in Tableau and was connected to the supporting census tables, including the 2021 health dataset and the 2011 economy, health, housing and qualification pivot tables. These relationships were defined as one-to-many, with the master table using as the main analytical layer for the dashboard.

3. Task Definition

This section explains the main analytical functions shaped by Munzner's task abstraction framework [1]. Rather than implementing the dashboard by putting just charts and graphs, Munzner's approach focuses on what user needs to do with the data. In this project, the main motivation is to analyse socio-economic features with general health distribution, compare regions and local authorities, locate spatial

patterns, identify unusual areas, discover clusters and understand uncertainty in model-based estimates.

The dashboard was designed for doing exploratory analysis on Census data rather than answering only one fixed question. Flow is like this: users can first get the overview of 2011 socio-economic features, then explore patterns between local authorities and regions using PCA and t-SNE projections, and finally compare health outcomes between the 2011 and 2021 Census. In addition, the Bayesian uncertainty view also supports users in assessing model-based predictions by showing lower and upper uncertainty bounds for predicted unhealthy population values. All of these tasks are linked to the main aim of the project, which is to understand regional socio-economic and health patterns in England and Wales.

Dashboard view / element	User task	Munzner's action	Target
2011 socio-economic overview	Understand overall socio-economic and health conditions	Query: Summarise	Key attributes / distributions
Regional charts and health comparison	Compare socio-economic and health values across regions and years	Query: Compare	Regional values / trends
Choropleth maps	Explore where high, low or changing values occur geographically	Search: Locate / Explore	Spatial patterns
PCA and t-SNE projection views	Discover similarities, clusters and possible outliers between local authorities	Analyse: Discover	Features / similarity / outliers
Bayesian uncertainty view	Assess uncertainty in predicted 2021 unhealthy population values	Query: Identify / Compare	Model-based estimates / uncertainty bounds

Table 1. Main analytical tasks supported by the dashboard using Munzner's task abstraction framework

4. Visualisation Justification

This section explains why these visualisation techniques were chosen for the dashboard. The main aim was to select the chart/ graph types that matches the data and tasks they needed to perform. Since the project includes socio-economic indicators, health categories, geographic areas, projection

coordinates and uncertainty values, it was challenging to justify all these aspects by using only single chart type.

4.1 Overall Design Approach

The dashboard was made in such a way that it should portray as visual analytics tool rather than fixed answer report.

- There are main three analytical views In the dashboard:
    - 2011 socio-economic overview
    - PCA and t-SNE projection view and their justifications
    - 2011–2021 health comparison and uncertainty view

- This layout allow user to move from overall view to broad detailed exploration.
- The design follows the idea of overview first, filter, then details-on-demand [2].
- Implementation of multiple types of visuals was necessary move because dataset itself contains varied datatypes, discrete and continuous variables, spatial data, projection coordinates and all.

4.2 Summary and Comparison techniques

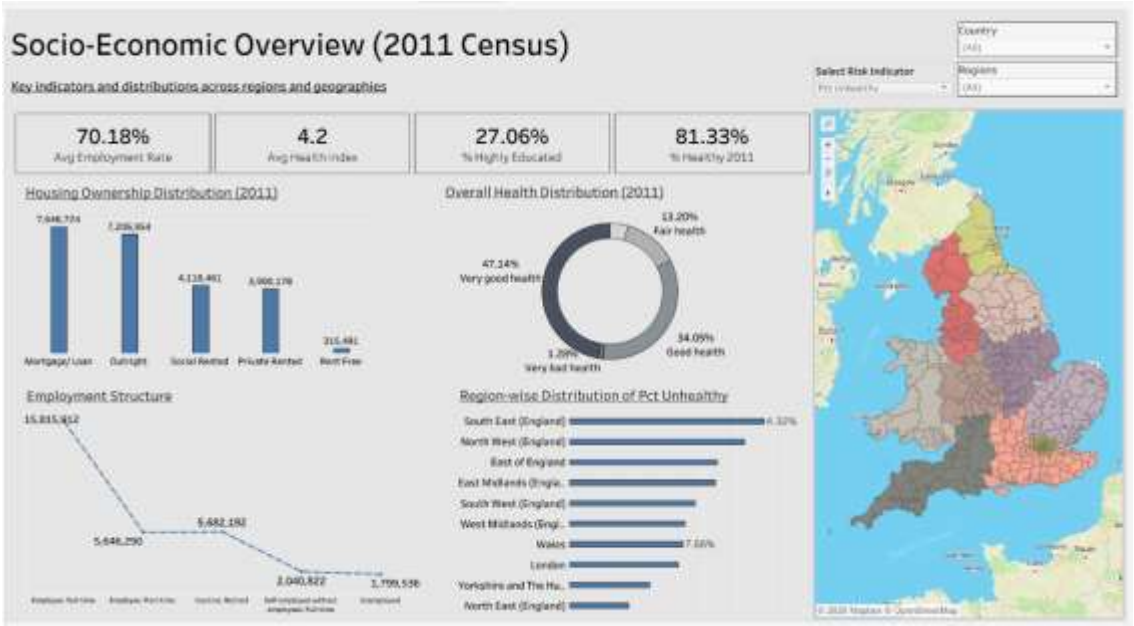


Fig.2. 2011 Socio-Economic Overview dashboard Overview

- KPI cards, quantitative charts and maps were used to support and summarize data.
- KPI cards show key values such as average employment rate, average health index, percentage highly educated and healthy population percentage. These cards give users a quick summarized overview of the overall data
- Bar and distribution charts were used to analyse how health and socio-economic categories are spread across the overall population.
- These techniques concisely reduce the cognitive overload by turning raw Census data into readable visual summaries.

4.3 Spatial Visualisation

- Choropleth maps were also used because data is Census data, which is highly linked with local authorities and regions.
- They help users locate where high, low or changing values appear geographically, and also work as interactive filters for exploring related charts.
- In the broad view of visual comparison of 2011 and 2021 health data (dashboard 3), map also adds spatial context to the change between census years, without relying on other visualisations.
- However, sometimes maps can be overstated on large rural areas, so other types of charts and KPIs are necessary for more precise comparison.

4.4 PCA and t-SNE Projections overview

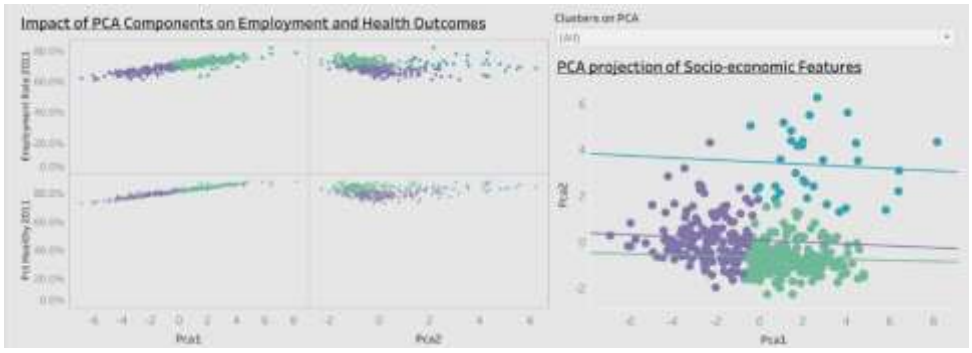


Fig. 3. PCA Projection Overview

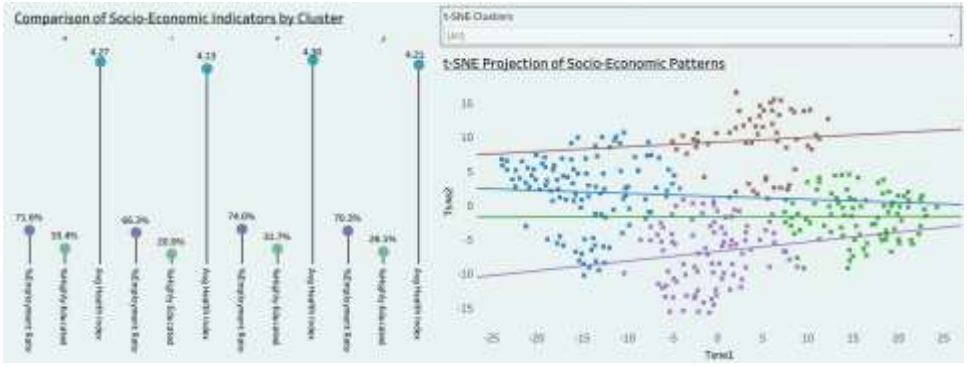


Fig.4. t-SNE Projection Overview

- PCA And t-SNE, these two dimensionality reduction techniques were used to reduce multiple socio-economic factors into two components.
- The input features were standardized before projection to avoid stretched scaling to dominate the result by any component.
- In the PCA view, PCA1 seems to show a clearer link with employment rate and the percentage of healthy population. This indicates that PCA1 reflects a broad socio-economic gradient, where areas with higher PCA1 values tend to have higher employment and better health outcomes. While PCA2 shows a weaker and less steady pattern, so it was taken as secondary variation rather than the main trend.
- t-SNE was also used to explore clusters between local authorities.
- As compared to PCA results, t-SNE showed more visible clusters, suggesting possible socio-economic similarity between some local authorities. However, feature wise comparison of clusters did not differ strongly or very subtle difference between them. Therefore, t-SNE clusters were treated as exploratory groups rather than features.
- PCA gives a more interpretable linear summary, while t-SNE should be interpreted more carefully because it focuses on local similarity rather than exact global distances.
- These projections were used for exploration and pattern discovery, not for proving causation.

4.5 Bayesian Uncertainty Visualisation



Fig.5: Bayesian Projection Overview

- Bayesian regression was used to estimate 2021 unhealthy population with credible intervals (lower-upper bonds)
- Error bar chart was used here to show uncertainty range between lower and upper credible values.
- Predicted values across local authorities seems very similar which denotes limited variation captured by the model.
- This pattern reflects Bayesian smoothing, where predictions follow overall socio-economic trends rather than strong local differences.

4.6 Interaction options in Visualisations

- Filters for country, regions, clusters and parameters allow user to interact with dashboards to retrieve specific results.
- Tooltips provide important details about charts (when user hover on the chart) without overcrowding the dashboard.
- PCA/t-SNE buttons control allow users to switch between projection methods to see the respective overall results.
- These interactions support exploratory analysis and reduce visual clutter and cognitive load.

4.7 Limitations

Although the dashboard is interactive and informative, there are some limitations that need to be considered.

- Maps might overemphasize large geographic areas.
- PCA and t-SNE projections are dimensionality reduction techniques at the end, so some details are lost.
- In Bayesian Uncertainty chart, the visual design apparently blurs distinction across local authorities because their bar lengths are very similar to each other. So, this chart might be less intuitive to some users who are unfamiliar with Bayesian principles.
- The dashboard shows patterns and associations, but it does not prove causation.

5. Evaluation

By considering Munzner's chapter 4 of Validation Framework, the evaluation primarily focused on confirming task and data abstraction, along with visual encoding and interaction patterns. The motivation was to examine whether the dashboard implemented the desired analytical activities, like summarizing socio-economic conditions, comparing areas, recognizing spatial patterns, analysing PCA/t-SNE groupings, and understanding Bayesian uncertainty. It also evaluated that whether implemented visual encodings and interactions such as KPI cards, bar charts, maps, filters, and tooltips, were useful clear to users.

5.1 Evaluation Method

The dashboard was evaluated during the practical evaluation session arranged by the University module. Five classmates from the discussion group reviewed the dashboard during this session. Although the evaluation took place in a formal class setting, and it was formative rather than a controlled usability experiment with time bound tasks.

The 2011 socio-economic overview, the PCA/t-SNE projection view, and the 2011–2021 health comparison with Bayesian uncertainty were the three primary dashboard views displayed during the session. Two students directly engaged with the dashboard by applying filters and examining

particular displays, however the dashboard was primarily displayed through a walkthrough. After looking over the dashboard, the other peer members provided the feedback.

5.2 Validation Focus

- Task and data abstraction validation was applied when considering how well the dashboard supported specific analysis tasks such as summarizing socio-economic characteristics, comparing different regions, discovering spatial trends, visualizing clusters in PCA/t-SNE outputs and understanding uncertainty.
- Visual encoding and interaction idiom validation was applied to determine whether the selected visual methods were perceivable and usable such as the KPI cards, bar chart, choropleth map, projection plot, filter and tooltip.
- Algorithm validation was less emphasized, being assessed primarily on the interpretability of the PCA/t-SNE output and Bayesian uncertainty within the dashboard's context.

So, the evaluation did not aim to produce statistical feedback. However, it gave formative comments on whether the dashboard design was intelligible and usable for exploratory analysis.

5.3 Observations and Feedback

The following table summarises the main observations and feedback provided by the students during the dashboard evaluation.

Evaluation aspect	Observation/ Feedback
Overall dashboard structure	The three-view structure was easy to follow, moving from 2011 overview, to projection analysis, and then to 2011–2021 health comparison.
KPI cards and comparison charts	These were understood quickly because they used familiar visual forms and gave a clear summary of key values.
Choropleth maps	The maps helped connect patterns with geography and were also useful as interactive filters.
PCA and t-SNE views	The projection views were visually interesting, but they needed some explanation so users understood that the points represented socio-economic similarity, not geographic location.
Bayesian uncertainty view	The uncertainty chart was useful but less intuitive than the standard charts
Interaction	Filters and tooltips supported exploration and helped users focus on specific regions, clusters or values.
Overall Feedback	The feedback was positive, especially regarding the dashboard layout, interactivity and visual clarity. No major confusion was observed during the walkthrough.

Table 2: Summary of Peer Evaluation Feedback

5.4 Evaluation Reflection

Overall, according to the evaluation, the dashboard was easy to use and beneficial for exploratory evaluation. Because KPI cards, bar charts, and maps employed well-known visual forms and facilitated rapid comparison, they were simple to understand.

Some explanation was needed for the more technical perspectives, particularly PCA/t-SNE and Bayesian uncertainty. Users had to comprehend that although Bayesian intervals indicated uncertainty around expected values, projection points demonstrated socio-economic similarity.

Despite these remarks, clear tooltip plus interactive hints could improve how projections and uncertainties appear. While the dashboard supported the intended work, guidance on complex parts would help users later. Though effective overall, some features may need simpler explanations going forward.

6. Conclusion

Overall, project analysed and thoroughly visualised the data from the 2011 and 2021 Censuses to examine specific socio-economic and health trends in England and Wales. Health and socio-economic situations are not equally spread among areas and local authorities, according to the report. Employment, education, housing, and self-reported health were among the indicators that provide helpful context for comprehending regional variations. While the 2011–2021 comparison demonstrated that health outcomes should be evaluated locally and regionally rather than solely through national averages, the 2011 overview assisted in establishing a socio-economic baseline. Although these categories should be regarded as exploratory rather than causal, the projection views also indicated that certain local authorities have comparable socio-economic profiles.

From the standpoint of information visualisation, this coursework demonstrated that tasks, not chart diversity, should be the driving force behind dashboard design. Maps facilitated spatial exploration, PCA and t-SNE revealed similarity patterns, KPI cards and bar charts were helpful for summary and comparison, and Bayesian uncertainty intervals prevented model-based estimations from being presented as exact values. The assessment also revealed that while technical views like projections and uncertainty charts required more explanation, familiar visual forms were simpler for users to comprehend. Overall, the experiment showed that when data preparation, visual encoding, interactivity, and uncertainty are planned together, visual analytics can help make complex census data easier to grasp.

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